NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

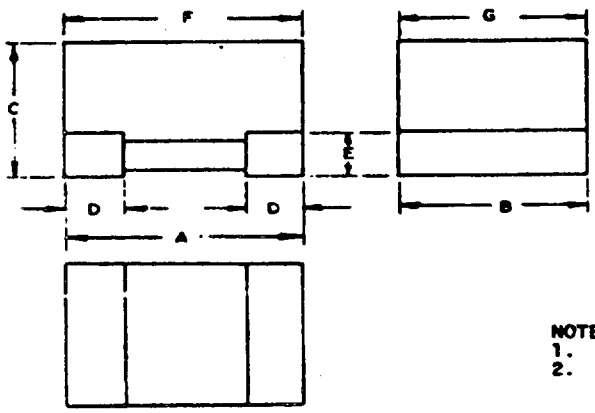
MIL-PRF-83446/1B
16 DECEMBER 1980
SUPERSEDING
MIL-I-83446/1A
4 May 1978

PERFORMANCE SPECIFICATION SHEET

COILS, CHIP, RADIO FREQUENCY, FIXED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the coils described herein shall consist of this document and the latest issue of Specification MIL-C-83446.



	INCHES	MM
A	.150 ±.005	3.81 ±.13
B	.120 ±.005	3.05 ±.13
C	.080 max	2.03 max
D	.025 ±.005	.64 ±.13
E	.035 max	.89 max
F	.155 max	3.94 max
G	.125 max	3.18 max

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Fixed, chip coil.

(B)

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 0.5 gram maximum.

Operating temperature range: -55° to +125°C.

Storage temperature range: -55° to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 700 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202, (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): See table I.

(B) denotes changes

FSC 5950

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(B) TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number	Inductance 1/	Q (min)	Q (typ)	2/ Frequency	Self resonant frequency (min)	DC resistance (max)	3/ Current (max)
	uH			MHz	MHz	Ohms	mA
01	0.010	48	58	150	900	0.050	650
02	0.012					0.055	
03	0.015					0.060	
04	0.018					0.065	
05	0.022			100		0.070	
06	0.027					0.075	
07	0.033					0.075	
08	0.039					0.080	
09	0.047				850	0.085	
10	0.056				800	0.088	
11	0.068				750	0.093	
12	0.082				700	0.095	
13	0.100	50	60	25	600	0.075	
14	0.120				550	0.075	
15	0.150				420	0.085	
16	0.180				390	0.10	650
17	0.220				340	0.11	640
18	0.270				290	0.12	620
19	0.330				230	0.14	590
20	0.390				210	0.16	550
21	0.470				190	0.18	520
22	0.560				180	0.20	500
23	0.680				170	0.23	460
24	0.820				150	0.26	430
25	1.00				140	0.34	380
26	1.20	36	45	7.90	130	0.42	370
27	1.50	36	45		120	0.56	320
28	1.80	36	45		100	0.76	280
29	2.20	36	45		98	0.93	250
30	2.70	40	50		91	1.2	220
31	3.30	40	50		76	1.3	210
32	3.90	47	60		68	1.5	200
33	4.70	47	60		46	1.7	180
34	5.60	44	55		42	1.8	170
35	6.80	40	50		39	1.9	160
36	8.20	40	50		30	2.4	150
37	10.0	46	58		26	3.2	130
38	12.0	41	52	2.50	24	3.7	120
39	15.0	46	58		23	3.8	110
40	18.0	46	60		22	4.2	100
41	22.0	47	60		18	5.5	98
42	27.0	47	60		17	6.1	95
43	33.0	47	60		13	6.6	92
44	39.0	50	62		12	7.0	88
45	47.0				11	8.3	85
46	56.0				10	8.9	82
47	68.0				9.1	13.0	67
48	82.0				8.6	14.0	65
49	100.0	47	60		7.6	16.0	61
50	120.0	30	38	0.790	6.8	17.0	59
51	150.0	32	40		5.6	18.0	57
52	180.0				4.5	22.0	52
53	220.0				4.0	26.0	46
54	270.0				3.8	32.0	43

See footnotes at end of table.

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(B) TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number	Inductance 1/	Q (min)	Q (typ)	2/ Frequency	Self resonant frequency (min)	DC resistance (max)	3/ Current (max)
	µH			MHz	MHz	Ohms	mA
55	330.0	32	40	0.790	3.5	44.0	37
56	390.0	32	40	↓	3.4	48.0	35
57	470.0	28	35	↓	3.2	75.0	28
58	560.0	↓	↓	↓	2.8	81.0	27
59	680.0	↓	↓	↓	2.7	93.0	23
60	820.0	↓	↓	↓	2.0	112.0	21
61	1000.0	↓	↓	↓	1.7	120.0	19

1/ Inductance tolerance: 0.01 to 0.082 µH ±20%.
0.10 to 0.0820 µH ±10%.
1 to 1,000 µH ±5%.

2/ Frequency range - 150 through 100 MHz:

Inductance shall be measured on instruments such as the HP260A, HP4342A, HP250B RX meter, or equivalent with Vanguard TF-18356-QMS test fixture, or equal (see figure 2). Fixture inductance (approximately 0.01 µH) and residual Q-meter inductance (0.0026 µH) should be subtracted from calculated inductance.

Frequency range - 25 through 0.79 MHz:

Inductance shall be measured on instruments such as the HP260A, HP4342A, HP250B RX meter, or equivalent with Vanguard TF-18356-QMB test fixture, or equal (see figure 3). Fixture inductance (approximately 0.025 µH) and residual Q-meter inductance (0.01 µH) should be subtracted from calculated inductance.

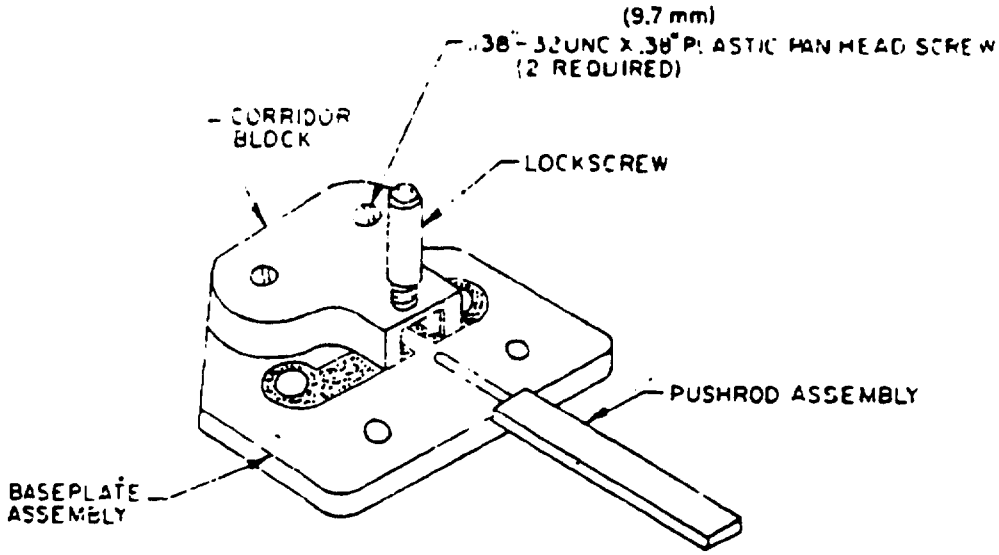
3/ Maximum current allowed not to exceed the specified temperature rise.

Electrical characteristics (final): See table II.

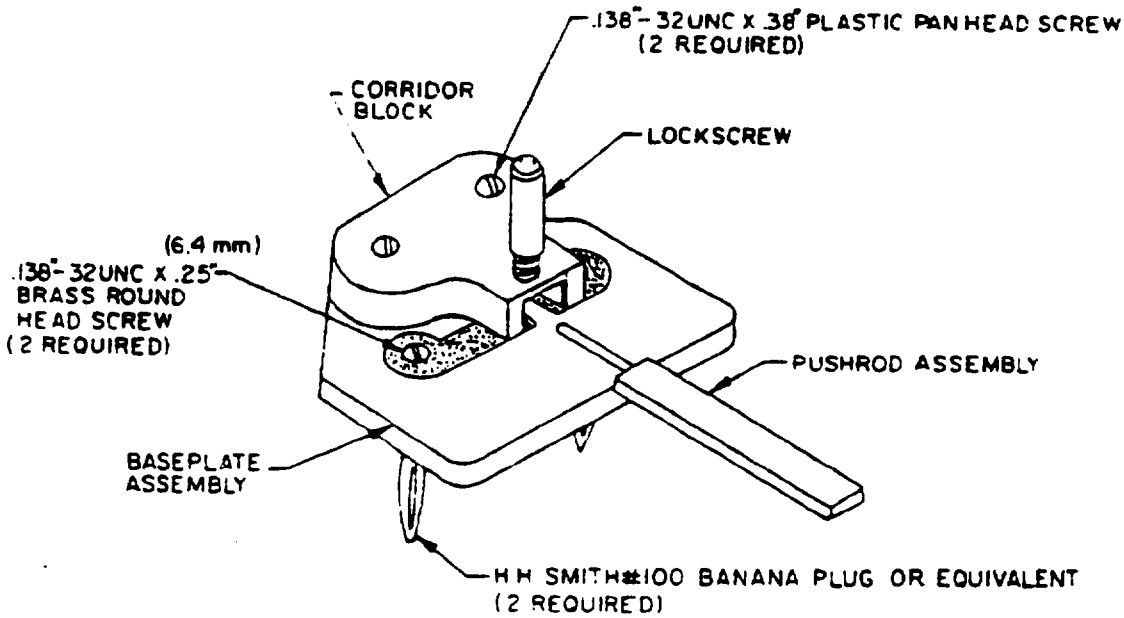
TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from the initial measurements			
	Inductance	DC resistance	Self-resonant frequency	Q
Qualification inspection	Percent		Percent	Percent
Group II	±5	±(3% +0.001 ohm)	-8	-10
Group IV	±5	±(2% +0.001 ohm)	-10	-10
Group V	±2	---	---	-10
Quality conformance inspection				
Group C				
Subgroup II	±5	±(3% +0.001 ohm)	-8	-10
Subgroup IV	±5	±(3% +0.001 ohm)	-8	-10

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Ⓑ FIGURE 2. Typical chip coil test fixture (TF-18356-QMS, or equal).



Ⓑ FIGURE 3. Typical chip coil test fixture (TF-18356-QMB, or equal).

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(E) Supersession data: A new part numbering system with codes for termination materials was incorporated with this revision, superseding MIL-I-83446/1A, dated 4 May 1978.

M83446/01- (dash number from table I) B
supersedes M83446/1- (dash number from table I).

Examples:

M83446/01-21B supersedes M83446/1-21.
M83446/01-03B supersedes M83446/1-3.

(B) Part number: The part number shall be in the following form.

M83446/01-	04	B
Sequentially assigned dash numbers (see table I)		
Termination finish (see MIL-C-83446)		

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS
Air Force - 19
DLA - ES

User activities:
Navy - OS
Air Force - 11

Preparing activity:
Army - ER

Agent:
DLA - ES

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